



The simplest and cheapest method of starting wound-rotor induction motors is by means of added rotor resistance, with full-line voltage across the stator terminals. At the time of start, the addition of external resistance in the rotor circuit of a wound-rotor induction motor i) decreases its starting current ii) increases its starting torque and iii) improves its starting power factor.

A wound-rotor induction motor is used for loads requiring large starting torque or for loads requiring speed control. A wound-rotor induction motor is more expensive than a squirrel-cage motor and also it requires more maintenance because of the brushes and slip rings. A wound-rotor motor may be used for hoists, cranes, elevators, compressors etc.

PROCEDURE

A) NO LOAD TEST

Make the connections as shown in the diagram.

Precautions :

- i) Keep the autotransformer in minimum voltage position
- ii) Keep belt on brake drum in loose position (motor on no load) in) Starter handle should be in maximum anticlockwise position (External rotor resistance is maximum)

Switch on the 3 phase supply. Adjust the autotransformer and apply rated voltage to the stator. By pressing the push button provided on the starter, rotate the handle of the starter slowly in clockwise direction so that rotor resistance is gradually cut off. Now, the motor runs on no-load. Since the power factor on no-load is quite low, less than 0.5, (the no load current is mainly used for magnetizing the core which is largely inductive in nature), one of the wattmeter will read negative. Then switch off the supply and interchange the connections of the pressure coil (or current coil) of that wattmeter and again start the motor by the above procedure. Note down the ammeter, voltmeter and wattmeter readings. The sum of the wattmeter readings shows the rotational losses (rotational losses = core loss + mechanical losses).